Aurum: Cleaner Share, More Fossil Power?

Milestone 2: Data Handling and Initial Sketches
CS-GY 6313 Information Visualization, NYU, Fall 2026 · November 2, 2026
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Repository: https://github.com/Yahtze/InfoViz-Aurum

Scope and Current State

This report covers **all six questions**: Q2–Q4, which Yathin explored with six Matplotlib sketches, and Q1, Q5, and Q6, which Yusuke explored with three. They are exploratory views, not the final D3 application. The detailed reports in the repository are dataHandlingQ2Q3Q4.md (Q2–Q4) and dataHandlingQ1Q5Q6.md (Q1, Q5, Q6).

The baseline is 2024 for the snapshot and 2000–2024 for comparisons, with a minimum of 10 TWh generation in 2024. The sketches show up to 10 countries per tail. This report reads the preparation baseline tables, not custom sketch settings. Custom sketch runs export their own CSVs and settings.

The tables below contain real processed rows, not synthetic examples. They are **case samples**, not random or population-weighted samples. Each section states the selection rule. Q4 contains all 10 plotted countries. The report displays two decimal places; downloadable CSVs retain four decimals where applicable. Rounding can make displayed sums differ by 0.01. **No Data** means unknown, not zero. CSV samples use blank cells for the same unknown values. Q5 and Q6 show associations, not causes.

Changes Since M1

> **We changed the comparison period and Q4. The problem, goal, hypothesis, dataset, and size threshold are the same as in the M1 proposal.**

	M1 proposal (Oct 19)	M2 (now)
Comparison period (Q3, Q5, Q6)	2010–2024	**2000–2024**
Q3	"…between **2010** and 2024?"	"…between **2000** and 2024?"
Q4	"…how much of the change came from **wind and solar compared with hydro and nuclear**?"	"…how much did **wind, solar, hydro, nuclear, and other clean sources each** contribute?"
Q6	"…the initial electricity mix in **2010**…"	"…the initial electricity mix in **2000**…"

Q1, Q2, and Q5 keep their M1 wording. Cutoffs and Sensitivity explains the move to 2000 and checks the main result for other start years and size thresholds.

Response to the M1 Feedback

Feedback	What we did in M2	Where
Point 1: choose Q4's measure	Both measures for the same 10 countries. They can disagree: Kenya's hydro lost 7.02 pp of share but generated 1.79 TWh more.	Q4
Point 2: define the terms	Clean (includes nuclear), fossil, share, pp, TWh, and demand.	Shared Data Handling

Feedback	What we did in M2	Where
Point 3: justify the cutoffs	Reasons for 2000 and 10 TWh, and a sensitivity check.	Cutoffs and Sensitivity
Direction 1: compare similar demand growth	Group comparison, a hypothesis check, and six matched pairs.	Q6
Direction 2: Δ clean share vs. Δ fossil TWh	A scatterplot with units on both axes and an enlarged ±60 TWh band.	Q5
Missing data, shared scales, cues beyond color	Unknown values are shown as "No Data", not as zero. Compared views share scales. Q1, Q3, Q5, and Q6 do not rely on color alone.	Q1–Q6

The repository is shared with `ctsilva`. The README gives the dataset URLs and the AI-use table.

Shared Data Handling

Sources

- Energy: pinned OWID energy file, data/raw/owid_energy_2026-09-30.csv (CC BY 4.0). Processing reads the local copy because the upstream download changes over time.
- Map: pinned world-atlas 2.0.2, data/raw/world-atlas-2.0.2-countries-50m.json (ISC; geometry from Natural Earth).

Definitions

- Clean electricity is nuclear, hydro, wind, solar, bioenergy, and other renewables excluding bioenergy. “Clean” is our short name for OWID's low-carbon category.
- Fossil electricity is coal + gas + oil. These are electricity measures, not total energy use.
- All six Q2–Q4 chart titles state **Domestic Electricity Generation**. These views do not measure all energy use or classify imported electricity by source.
- Shares use domestic electricity **generation**, not demand, as the denominator.
- Percentage points (pp) measure share differences: 20% to 30% is +10 pp, not +10% relative growth.

Filters and Cleaning

1. Retain 2000–2024, inclusive. The preparation code excludes 2025 as incomplete.
2. Keep country/territory rows with three-letter ISO codes. Assign XKX to Kosovo. Exclude regional aggregates and World from the Q2–Q4 selections.
3. Fill a blank source TWh with zero only when the known source TWh already sum to total generation within 0.01 TWh. Do not fill unknown totals or clean shares.
4. After that completeness check, fill a missing source share as 100 × source TWh / generation TWh, only if generation is positive. Shares remain unknown for zero generation.
5. For endpoint comparisons, join by ISO3. Require known clean shares and fossil generation in both years and at least 10 TWh generation in 2024. The baseline has **108 eligible countries**. Q2 has no size threshold.
6. Map ISO3 to three-character ISO numeric IDs with pycountry. Preserve leading zeros such as Algeria's 012. The on_map flag tests atlas country-ID membership. The map reads only country geometry, preserves holes, and handles antimeridian crossings.
7. For sketches, break cutoff ties by country name and show at most 10 per tail. Q3 gain/loss tails exclude zero changes. The stored comparison rank is a descending row rank; its equal-value order does not specify the sketch's alphabetical tie policy.

Additional Definitions

- TWh (terawatt-hours) measures amounts of electricity: 1 TWh = 1 billion kWh.

- Demand is domestic generation plus net imports (OWID electricity_demand_twh). Generation and demand differ when a country trades electricity: 29 of the 108 compared countries had net trade above 10% of their 2024 demand (15 net importers, 14 net exporters).

Stored Tables

File in data/processed/	One Row Per	Size	Uses
country_year.csv	Country or World × observed year	5,518 × 31	Q1 (World row); Q2; Q3–Q6 endpoint/source inputs
comparison_2000_2024.csv	Eligible country	108 × 46	Q3–Q4; Q5–Q6 use the same 108 countries
robustness.csv	Start year × size threshold	20 × 7	Cutoffs and Sensitivity

Country-years are observed records, not a fully imputed country × year grid. The in_comparison_set flag repeats fixed country membership across years; it is not a test of each individual year's eligibility.

Q1 — The World's Electricity Mix

Explored by Yusuke.

Question: How has the global electricity mix changed from 2000 to 2024, in shares and in TWh?

Transformation: select the World row for 2000–2024. Group the sources into Coal, Gas, Oil, Wind, Solar, Hydro, Nuclear, and Other Clean. The sketch checks that each year's values add up to total generation and to 100% within 0.01.

Result: the clean share rose from 35.3% to 40.9%, but total generation grew from 15,279 TWh to 30,930 TWh. So fossil generation also grew, from 9,879 TWh to 18,276 TWh. Q5 asks the same question for each country.

Sample: every fourth year from 2000 through 2024, in shares. q1_world_twh_sample.csv has the same years in TWh.

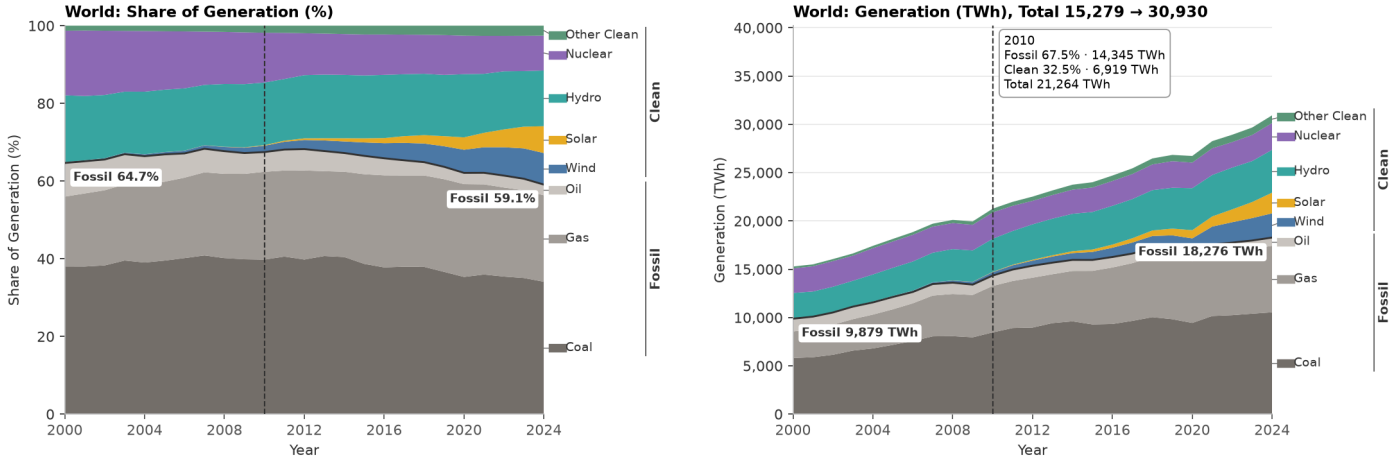
Year	Coal (%)	Gas (%)	Oil (%)	Wind (%)	Solar (%)	Hydro (%)	Nuclear (%)	Other Clean (%)	Total Generation (TWh)
2000	38.02	17.97	8.67	0.20	0.01	17.21	16.63	1.29	15,278.78
2004	39.03	20.12	7.26	0.49	0.02	16.09	15.62	1.38	17,414.99
2008	40.16	21.69	5.83	1.10	0.06	16.14	13.40	1.61	20,102.64
2012	39.80	22.92	5.50	2.35	0.43	16.26	10.80	1.94	22,515.66
2016	37.77	23.69	4.39	3.89	1.33	16.26	10.41	2.26	24,694.51
2020	35.31	23.89	2.90	5.96	3.19	16.27	9.91	2.57	26,723.37
2024	34.07	22.25	2.76	8.11	6.93	14.33	8.98	2.56	30,930.23

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Sketch: two stacked area charts with the same eight layers, in shares (%) and in TWh. Fossil layers are grays at the bottom, and a dark line marks their top. Clean layers use the Q4 colors. Brackets label the Fossil and Clean groups, so the split does not rely on color alone. The top of the TWh stack is total generation. **Planned interaction:** a cursor marks the year in both charts, and a tooltip lists every source. Clicking a source's name shows it alone as a line in both units, because middle layers are hard to compare in a stack.

Domestic Electricity Generation: World Mix by Source · 2000–2024

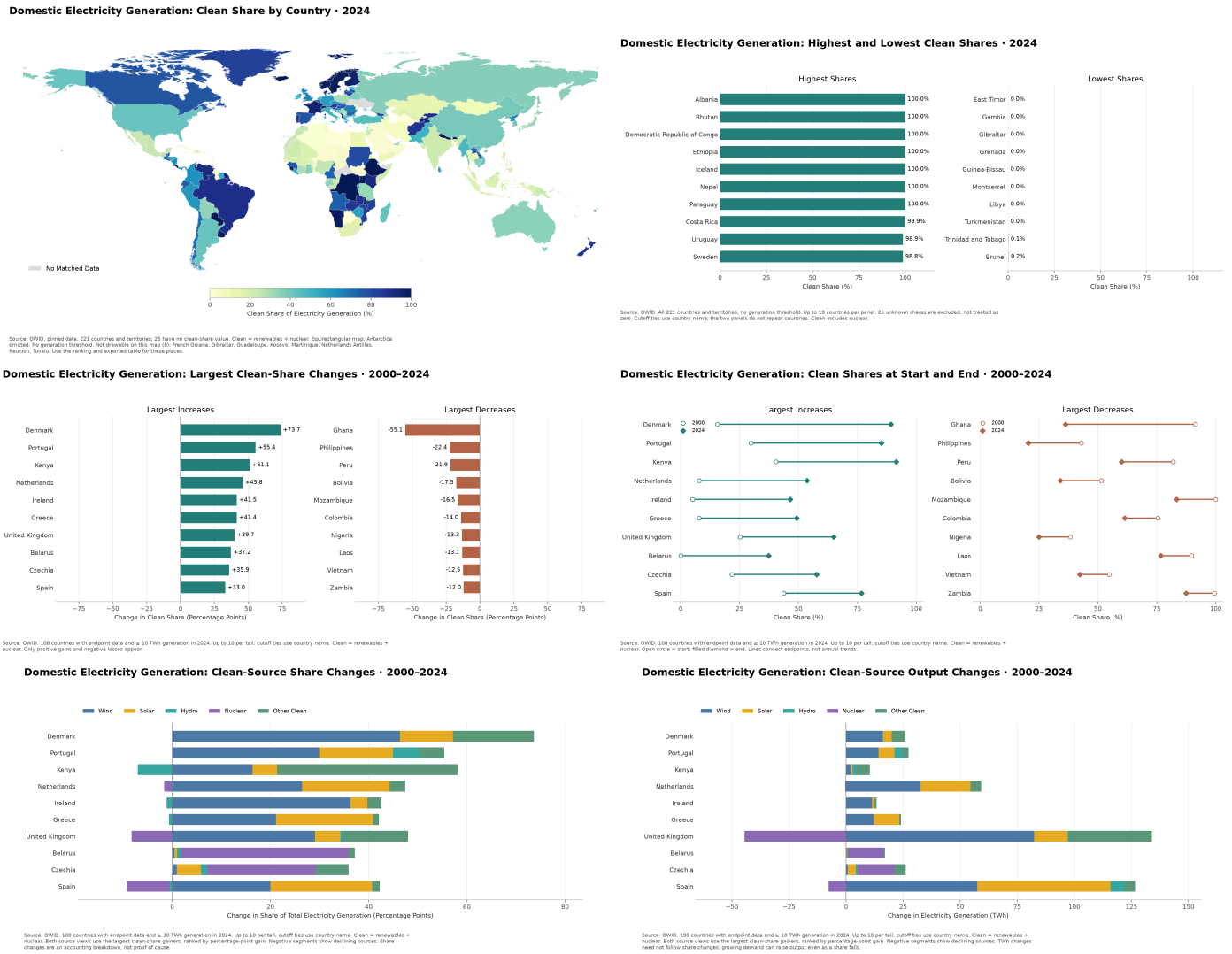
Planned Interaction: move the mouse across either chart; a cursor marks the year in both and a tooltip lists every source's share and TWh. Click a source's name at the right to show that source alone as a line in both units. Shown: the cursor at 2010 (tooltip shortened).



Source: OWID, pinned data. World row, 2000–2024. Shares are of domestic generation; the 8 layers add up to total generation and to 100%. Fossil sources are the grays at the bottom, and the dark line is their top; clean sources use the Q4 colors. Clean = renewables + nuclear; Other Clean = bioenergy + other renewables excluding bioenergy.

Q2 — Country Clean Electricity Shares

Explored by Yathin.



The image above holds the six Q2–Q4 sketches: two views per question.

Question: How do countries compare in their clean shares in 2024? Which have the highest and lowest shares?

Transformation: select year 2024 and country/territory rows, excluding World. Keep countries of every generation size. Sort known clean shares for the ranked tails. Join numeric IDs to atlas geometry, but retain unmapped places in the tabular data.

Coverage: 221 countries and territories, including 25 with unknown clean shares. 8 places have no drawable map geometry. A false map flag does not imply a missing electricity value.

Sample: three countries at 100%, three at 0%, three large generators, and three cases of map coverage or missing values.

Map IDs are text; keep leading zeros when importing the CSV.

Country	ISO3	Map ID	Total Generation 2024 (TWh)	Clean Share 2024 (%)	On Map
Albania	ALB	008	8.34	100.00	Yes
Bhutan	BTN	064	11.00	100.00	Yes
Democratic Republic of Congo	COD	180	15.92	100.00	Yes
East Timor	TLS	626	0.51	0.00	Yes
Gambia	GMB	270	0.51	0.00	Yes

Country	ISO3	Map ID	Total Generation 2024 (TWh)	Clean Share 2024 (%)	On Map
Gibraltar	GIB	292	0.22	0.00	No
China	CHN	156	10,086.88	38.17	Yes
Germany	DEU	276	495.99	58.64	Yes
United States	USA	840	4,391.02	41.87	Yes
Kosovo	XKX	No Data	6.77	8.57	No
Ukraine	UKR	804	No Data	No Data	Yes
Lesotho	LSO	426	0.00	No Data	Yes

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Sketch A — Geographic View: a world choropleth with a common 0–100% color scale. Gray means no matched data. The equirectangular map is not equal-area; Antarctica is omitted. **Sketch B — Ranked View:** horizontal bars for the 10 highest and 10 lowest known shares. Unknown values are excluded, not replaced with zero. Several leaders tie at 100%, so the cutoff does not establish a unique best country.

Q3 — Largest Changes in Clean Share

Explored by Yathin.

Question: Which countries had the largest increases or decreases from 2000 to 2024?

Transformation: use the 108 eligible endpoint pairs. Calculate d_clean_share_pp = clean_share_pct_2024 – clean_share_pct_2000. Select the top 10 strictly positive changes and the top 10 strictly negative changes separately. This is a percentage-point difference, not percentage growth.

Sample: the three largest gains, the three largest losses, and China and India for reference.

Gain Rank refers to all 108 compared countries, not to a separate loss ranking.

Country	Gain Rank	Clean Share 2000 (%)	Clean Share 2024 (%)	Clean Share Change (pp)
Denmark	1	15.54	89.19	73.65
Portugal	2	29.78	85.19	55.41
Kenya	3	40.37	91.48	51.11
Ghana	108	91.35	36.22	-55.14
Philippines	107	42.89	20.47	-22.42
Peru	106	81.98	60.11	-21.87
China	24	17.87	38.17	20.29
India	48	16.81	22.49	5.69

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Sketch A — Change Magnitudes: horizontal gain/loss bars with a shared zero-centered scale. **Sketch B — Start and End Levels:** open circles mark 2000; filled diamonds mark 2024. Both sketches use the same countries and order. Lines join endpoints, not annual trends. This separates the size of a change from the starting and final electricity mix.

Q4 — Clean Sources Behind the Largest Share Gains

Explored by Yathin.

Question: Among the largest clean-share gainers, how did wind, solar, hydro, nuclear, and other clean sources each contribute to share and generation changes?

Selection: the top 10 positive clean-share gainers, ranked by pp, from the same eligible comparison set. Both views use the same countries and order. The TWh view does not reselect countries by absolute-output growth.

Transformation: subtract each source's start value from its end value. Keep Wind, Solar, Hydro, and Nuclear separate. Sum bioenergy and other renewables excluding bioenergy into Other Clean. Fossil sources are excluded. These five components partition all six clean sources. Their sum equals the clean-share change in pp or clean-generation change in TWh. Each share endpoint has its own total-generation denominator.

Sample A — Source Share Changes

Country	Wind Change (pp)	Solar Change (pp)	Hydro Change (pp)	Nuclear Change (pp)	Other Clean Change (pp)	Net Clean Change (pp)
Denmark	46.39	10.78	-0.03	0.00	16.51	73.65
Portugal	29.94	15.00	5.30	0.00	5.16	55.41
Kenya	16.36	4.98	-7.02	0.00	36.78	51.11
Netherlands	26.43	17.83	-0.09	-1.54	3.20	45.82
Ireland	36.27	3.51	-1.12	0.00	2.84	41.49
Greece	21.14	19.73	-0.70	0.00	1.24	41.41
United Kingdom	29.06	5.20	0.68	-8.27	13.08	39.74
Belarus	0.44	0.57	0.76	34.24	1.20	37.20
Czechia	0.92	4.93	1.24	22.15	6.68	35.91
Spain	20.00	20.74	-0.54	-8.75	1.53	32.97

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Sample B — Source Generation Changes

The same countries in TWh: Download CSV.

Sketch A: diverging stacked changes in share, in pp. **Sketch B:** diverging stacked generation changes, in TWh. Right-hand segments show increases; left-hand segments show declines. The plots show source contributions only, without net-change diamonds. The sample tables retain the signed net clean change for reference.

For example, UK wind rises by 82.34 TWh and solar by 14.79 TWh; hydro rises by 0.67 TWh, while nuclear declines by 44.47 TWh. Other Clean increases by 36.21 TWh. These five source contributions sum to net clean growth of 89.54 TWh; the separate values make the nuclear decline visible rather than masking it within a broader category.

This is an accounting breakdown, not proof of cause. A source can gain TWh but lose share when total generation grows faster. Missing source values cause the sketch calculation to fail rather than being treated as zero.

Q5 — Clean Share and Fossil Generation

Explored by Yusuke.

Question: Which countries raised their clean share while cutting fossil generation, and which raised it but still generated more fossil electricity?

Transformation: use the 108 countries of Q3–Q4. Calculate d_fossil_twh = fossil_twh_2024 – fossil_twh_2000. Sort each country into one of four outcomes: clean share up with fossil down, up, or unchanged, or clean share not up. Fossil change stays in TWh, because 4 countries had no fossil generation in 2000.

Result: of the 79 countries whose clean share rose, 35 cut fossil generation and 44 (55.7%) still generated more.

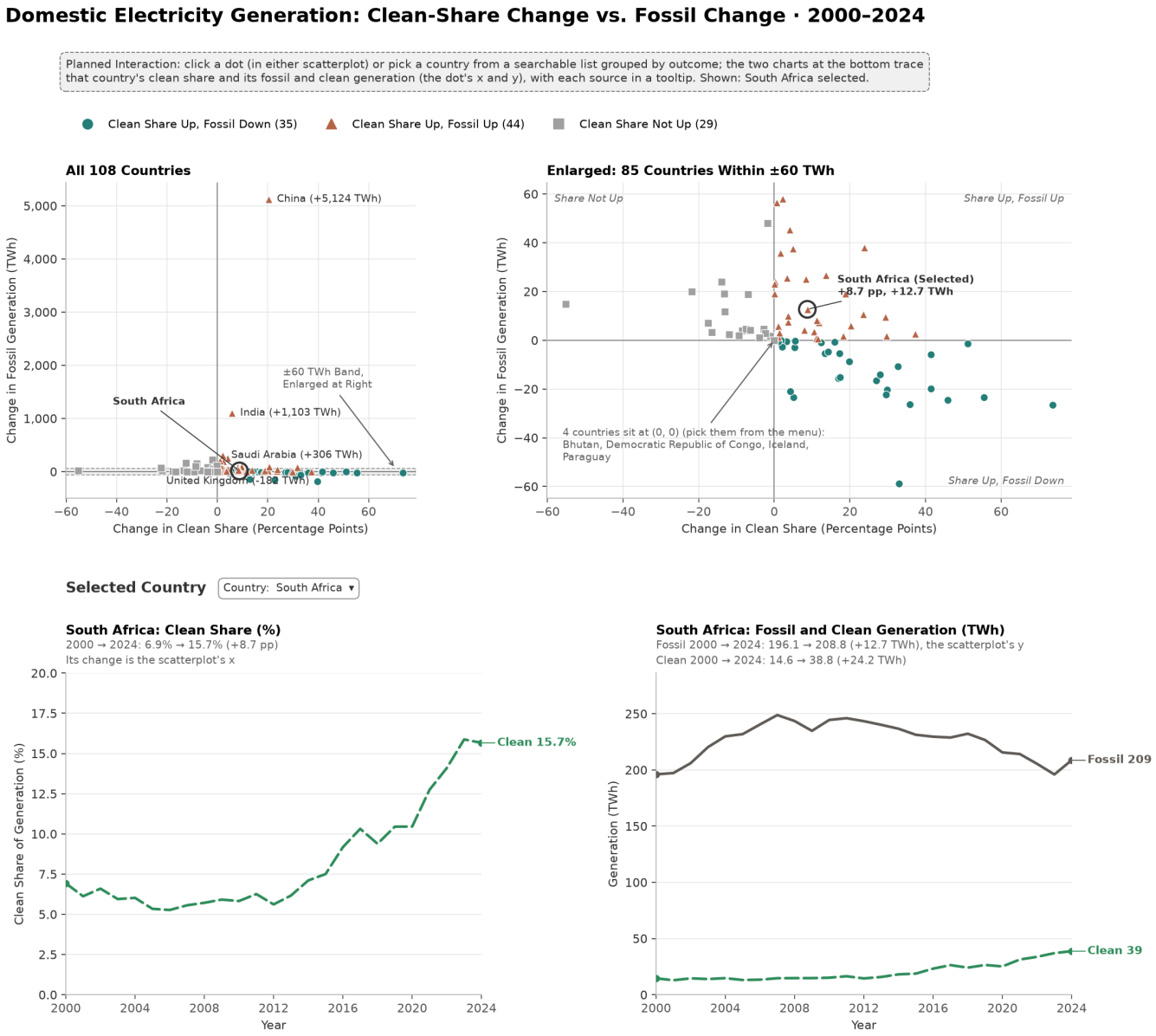
Sample: the 12 largest 2024 generators, plus South Africa. The sketch selects South Africa as the largest 2024 generator among the "share up, fossil up" countries inside the ±60 TWh band.

Country	Outcome	Clean Share Change (pp)	Fossil Generation 2000 (TWh)	Fossil Generation 2024 (TWh)	Fossil Generation Change (TWh)	Total Generation 2024 (TWh)
China	Clean Share Up, Fossil Up	20.29	1,113.30	6,237.20	5,123.90	10,086.88
United States	Clean Share Up, Fossil Down	12.81	2,697.28	2,552.53	-144.75	4,391.02
India	Clean Share Up, Fossil Up	5.69	475.35	1,578.43	1,103.08	2,036.52
Russia	Clean Share Up, Fossil Up	2.34	582.89	774.74	191.85	1,209.31
Japan	Clean Share Not Up	-6.92	679.52	698.37	18.85	1,016.39
Brazil	Clean Share Not Up	-1.78	30.62	78.68	48.06	745.72
Canada	Clean Share Up, Fossil Down	5.11	165.57	142.04	-23.53	636.79
South Korea	Clean Share Up, Fossil Up	0.96	177.37	375.91	198.54	625.38
France	Clean Share Up, Fossil Down	4.21	49.57	28.66	-20.91	561.78
Germany	Clean Share Up, Fossil Down	22.61	363.93	205.13	-158.80	495.99
Saudi Arabia	Clean Share Up, Fossil Up	2.16	138.69	444.82	306.13	454.64
Iran	Clean Share Up, Fossil Up	3.97	115.53	367.39	251.86	395.79
South Africa	Clean Share Up, Fossil Up	8.74	196.07	208.75	12.68	247.52

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Sketch: two scatterplots of clean-share change (x, pp) and fossil change (y, TWh): all countries, and an enlarged ±60 TWh band with the same x scale. The band holds 85 of the 108 countries, which overlap near zero next to China's +5,123.9 TWh. Color and marker shape both show the outcome. Below them, two line charts trace the selected country over 2000–2024: its clean share, and its fossil (solid gray) and clean (dashed green) generation. **Planned**

interaction: clicking a dot or choosing a country from a menu redraws the lines, and a tooltip lists each source by year.



Source: OWID. 108 countries with endpoint data and ≥ 10 TWh generation in 2024. Both scatterplots use the same linear x scale. Percent change in fossil generation is not used: small bases give extreme percentages, and 4 countries had no fossil generation in 2000. Example rule: the largest 2024 generator among "share up, fossil up" countries in the band. Changes are computed before rounding. Clean = renewables + nuclear. Not causal.

Q6 — Demand Growth and the Starting Mix

Explored by Yusuke.

Question: How do demand growth and the 2000 mix differ between countries that reduced fossil generation and those that did not?

Transformation: start from the Q5 table. Leave out the 4 countries with no fossil generation in either year, which leaves 104. Calculate demand_growth_pct = 100 × (demand_2024 / demand_2000 − 1). Group countries by fossil_reduced (d_fossil_twh < 0).

Sample A — Groups: demand growth by group.

Group	Countries	Median Demand Growth (%)	Lower Quartile (%)	Upper Quartile (%)
Fossil Generation Fell	35	10.41	2.80	30.35
Did Not Fall	69	165.27	79.25	231.50

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Starting mix: where fossil generation fell, the median 2000 clean share was higher: 51.3% against 23.9%. The middle halves overlap, so the starting share alone does not separate the groups. q6_start_mix.csv gives the medians of the four 2000 shares; separate medians need not add up to 100%.

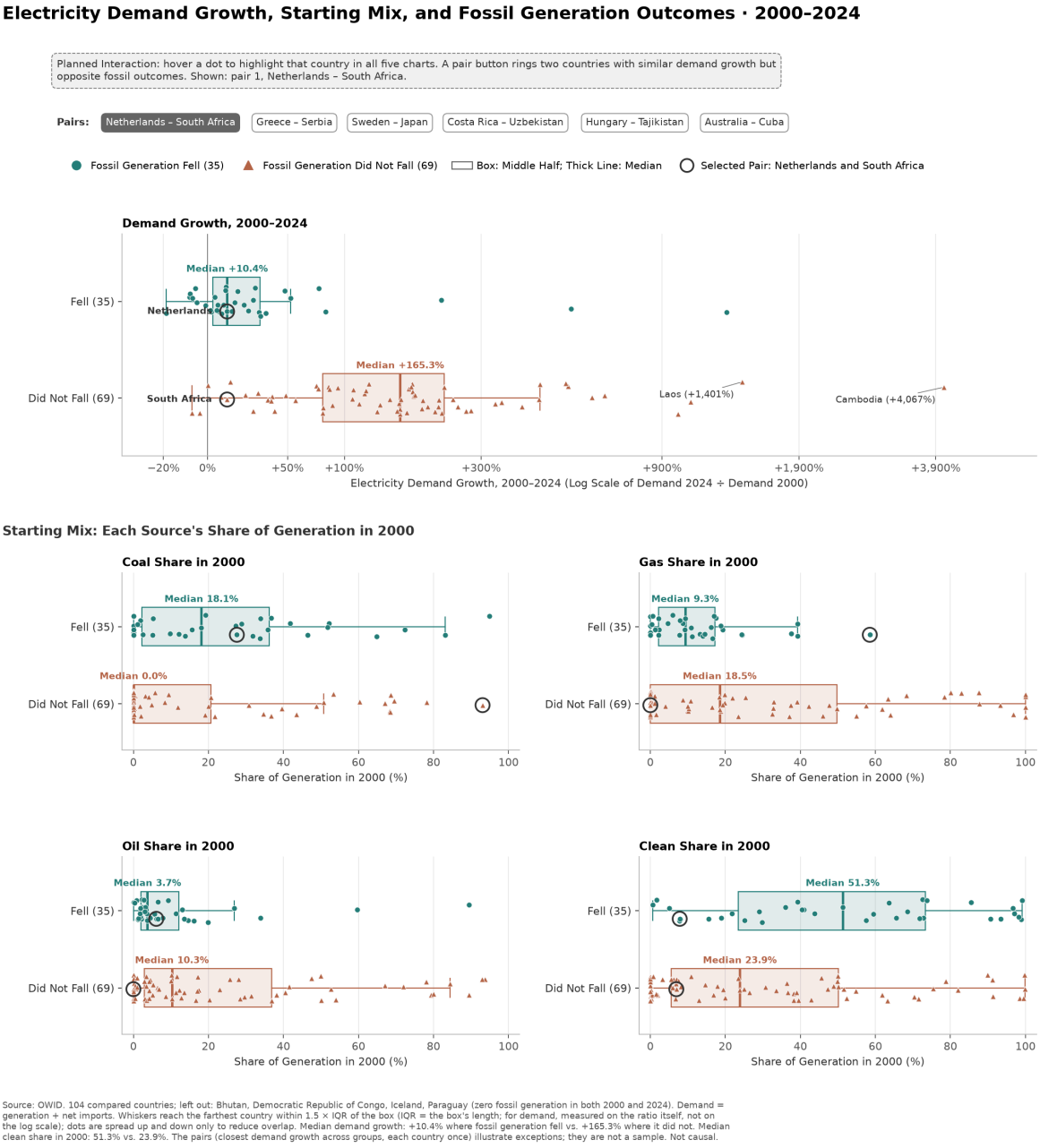
Hypothesis check: the proposal expected fossil cuts to be less common where demand grew fast; more frequent cuts there would contradict it. Among the 79 countries whose clean share rose, split at their median demand growth (75.4%), fossil generation fell in 30 of the 40 at or below it and in 5 of the 39 above it. This is consistent with the hypothesis but does not show a cause. The fast-growing exceptions are Costa Rica, Ethiopia, Kenya, Nepal, and Uruguay.

Sample B — Matched Pairs: for each country whose fossil generation fell, find the country with the closest demand growth whose fossil generation did not fall. Take the smallest gaps first, use each country once, and break ties by name. The 6 pairs illustrate exceptions, not a sample.

Pair	Fossil Fell	Demand Growth, Fell (%)	Fossil Did Not Fall	Demand Growth, Did Not Fall (%)	Gap (pp)
1	Netherlands	10.41	South Africa	10.42	0.00
2	Greece	7.25	Serbia	7.21	0.05
3	Sweden	-7.47	Japan	-7.57	0.10
4	Costa Rica	75.59	Uzbekistan	75.38	0.20
5	Hungary	26.15	Tajikistan	25.88	0.26
6	Australia	29.89	Cuba	29.27	0.61

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Sketch: five charts, each with one box per group (quartiles and median) and every country as a dot: demand growth, then the 2000 share of coal, gas, oil, and clean generation on a common 0–100% scale. Demand growth uses a log scale of demand_2024 / demand_2000 (1× = no change), labeled as percent change, because Cambodia's +4,066.7% would crowd the rest. A country's dot keeps the same height within its group in every chart. **Planned interaction:** hovering a dot highlights that country in all five charts; a pair button rings both countries of the pair.



Cutoffs and Sensitivity

Yusuke's analysis.

Start year 2000: the proposal started in 2010. The comparisons now start in 2000, like Q1. At 10 TWh, both start years compare 108 countries. An earlier start would compare fewer: the raw OWID file has generation data for 81 places in 1999 and 211 in 2000.

Size threshold 10 TWh in 2024: in a small system, one new plant can move the clean share by many points. The threshold keeps 108 of the 193 countries with data in both years.

Sensitivity: data/processed/robustness.csv recalculates the Q5 result for four start years and five thresholds: the percentage of countries with a rising clean share that also generated more fossil electricity.

Start Year	≥ 0 TWh (%)	≥ 5 TWh (%)	≥ 10 TWh (%)	≥ 20 TWh (%)	≥ 50 TWh (%)
2000	59.90	56.40	55.70	56.90	53.70
2005	52.50	48.90	49.40	50.00	50.00
2010	51.40	47.50	49.40	47.10	52.20
2015	46.70	45.70	46.70	45.80	47.90

Result: every setting has at least 20 countries whose clean share and fossil generation both rose, so the gap between share and amount appears with every cutoff. Whether they are a majority depends on the cutoffs. The proposal's 2010 start at 10 TWh gives 49.4% (41 of 83). The grid checks only the Q5 result, not Q6.

Created and Derived Features

Feature dictionary: **46 stored columns** used for Q2–Q4, with units, input fields, and transformation rules. It covers the relevant fields in the two processed tables, not unrelated fields. Copied/renamed source measures are distinguished from newly derived features.

Feature dictionary: **14 columns** that Q1, Q5, and Q6 use beyond the Q2–Q4 feature dictionary, with units, input fields, and transformation rules. Generation, the clean share, and the clean sources are documented there.

Sketch exports use _start and _end suffixes rather than year suffixes. Their source tables use configurable group labels as columns and total for net clean change. Custom sketch groups must include every clean source exactly once.

Reproduction and Checks

Run from the repository root:

```
uv sync --locked
uv run python prep/report_q234.py
uv run python -m sketches --preview outputs/sketchesQ2-4.png
uv run python -m prep.report_q156
uv run python -m unittest discover -s tests -v
```

- prep/report_q234.py checks source totals, clean shares, endpoint consistency, map coverage, finite values, and the main country counts.
- sketches/sketchesQ2Q3Q4/data.py recalculates selections and source changes for configured dates. Source groups must sum to net change within 0.01 in the relevant unit.
- The overview montage is tracked at outputs/sketchesQ2-4.png.
- sketches/q156_data.py checks that the eight Q1 groups add up to generation and to 100%, and recalculates the Q5 outcomes and the Q6 groups and pairs for configured dates.
- prep/report_q156.py checks that the documented fields exist. tests/test_q156.py checks that the Q1, Q5, and Q6 report's tables match what the default sketches compute.
- The figures are tracked at outputs/sketchesQ1.png, sketchesQ5.png, and sketchesQ6.png.
- Processed values already use four decimals. The report only rounds for display. It does not edit raw data or impute missing values.

AI Use

Date	Tool	What it produced	How we checked it
2026-10-18 to 11-02	Claude and Claude Code (Anthropic), Codex (OpenAI)	The M1 proposal draft; the repository setup and `prep/report_q234.py`; Q1, Q5, and Q6 code, report, and tests, based on designs chosen by Yusuke; the combined M2 report. Codex reviewed them	Yusuke reviewed the text and figures; tests check sums, counts, and that the report matches the sketches
2026-10-31	Codex (OpenAI)	Python code and Matplotlib syntax for Q2–Q4, based on plot types and visualization descriptions specified by Yathin	Automated tests check data selection, source totals, and plot structure